

Rhetorical Cohesion: A Multidimensional Measure of Party Unity in Political Messaging

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Abstract

Party unity is a central concept in legislative politics. We know voting cohesion is at record highs, but have no comparable understanding of rhetorical cohesion: whether co-partisans say the same things about the same issues, which is key to party branding and democratic accountability. Scholars have measured isolated dimensions of party talk, like ideology and partisanship, but party messaging is multidimensional. To address this gap, we develop a multidimensional measure of rhetorical cohesion using topic models and contextual embeddings. We validate our measure against human judgment and a well-known historical case. The measure exhibits theoretically-anticipated relationships with electoral timing, majority control, issue salience, and seat safety. By contrast, unidimensional measures yield inconsistent results because individual dimensions of party talk can move in different directions even when overall cohesion is rising. Our measure allows scholars to study the rhetorical side of party unity and better understand how party brands are built.

Word Count: 9,903

On June 13, 2024, then-Candidate Trump visited Capitol Hill—for the first time since the events of January 6, 2021—to rally congressional Republicans ahead of the presidential election. Although some co-partisan lawmakers expressed hesitation about the visit, Trump emerged from the meeting touting the party’s “great unity,” saying “I’m with them, a 1,000 percent. They’re with me a 1,000 percent. We agree just about on everything.” Although the circumstances of the visit were unusual, the emphasis on party unity was not. Political leaders routinely invoke unity as a valuable political commodity, yet they rarely specify what they mean by the term—treating it as a global property that yields positive benefits. As scholars, we need to operationalize and evaluate this construct, but existing measures may only capture it in parts.

Political scientists have sought to define and measure party unity more precisely. Normatively, advocates of Responsible Party Government argue that policy is made collectively, and parties are needed to collectively hold policymakers to account (Aldrich 1995; Schattschneider 1942). However, parties are necessary, not sufficient. As Fiorina (1980, 26, emphasis added) writes, “responsibility requires *cohesive* parties.” Further, empirical scholars argue that party members value unity. Individual lawmakers’ electoral fortunes are tied to the party’s collective reputation (Aldrich 1995; Cox and McCubbins 2005; Groeling 2010; Lee 2009). Given the importance of policy to party branding, scholars have operationalized unity through the roll-call record, and legislative voting serves as our primary indicator of party cohesion.

By roll-call measures, party unity truly is close to 1,000 percent. Party unity scores have never been higher (Reynolds and Maehr 2024), and parties have never been more polarized according to NOMINATE-based metrics.¹ However, these statistics cannot speak to the full breadth of party unity as advanced by normative theorists. As the APSA (1950,

¹Some have argued that these roll-call measures can be misleading. Lee (2018, 1464–1465) argues “roll-call voting data exaggerate party unity, especially in the contemporary era” where divisive issues are kept off the floor and where party messaging bills proliferate. Further, the increasing use of omnibus vehicles and unorthodox procedures leave rank-and-file members with little input or voting leeway (Curry and Lee 2020; Sinclair 2017), mechanically inflating roll-call unity scores.

17-18) report outlines, responsible parties must first “bring forth programs to which they commit themselves and, second...possess sufficient internal cohesion to carry out these programs.” Just as a party brand is strengthened when lawmakers vote as a partisan bloc, when lawmakers speak with one voice, they increase the likelihood that voters will internalize what they stand for, who they represent, and the standards by which they should be held accountable (Hopkins 2018; Lee 2016; Sellers 2009). If co-partisans promote different messages about the same policy, voters cannot know what a party stands for and cannot hold it accountable for the promises it has made. Roll-call voting provides insight into the second factor: carrying out programs. But votes cannot tell us whether parties publicly commit to programs.

Here, we develop a method to measure the program articulation side of party unity. We define our construct as *rhetorical cohesion*—the degree to which co-partisans say the same things about the same issues, which we operationalize as within-topic rhetorical similarity between party member pairs across several dimensions of party talk. This definition is intentionally broad, capturing alignment in positioning, framing, justification, tone, and other features of rhetoric that shape voter perceptions.² However, a speech, unlike an up-or-down vote, is multidimensional—which requires a purpose-built, multidimensional measure. Existing measures in the literature can capture single dimensions, such as ideology (Gaynor et al. 2025; Rheault and Cochrane 2020), national orientation (Ban and Kaslovsky 2024; Grimmer 2013; Noble 2024), or direct phrase reuse (Gaynor 2025), and they teach us about the causes and consequences of each independent element. Yet each targets a different dimension of party talk, and as we show, they can disagree when applied to the same question about party cohesion. Consider two lawmakers who oppose a health care bill but justify their opposition on different grounds. One argues it costs too much; the other argues it is a partisan bill designed to help the president win re-election. These two lawmakers vote the same way, share the same posi-

²As we discuss later, we treat rhetorical cohesion as a measurable output stemming from several forces, such as leadership pressure, common cause, selection into speaking, and bottom-up coordination.

tion, and hold the same ideological ground, yet the logic for their vote differs. A measure of ideology may register high similarity, but a measure of rhetorical cohesion *should not*. Therefore, we need a new measure that can simultaneously account for these (and other) dimensions, producing a summary statistic of rhetorical cohesion.

Our rhetorical cohesion workflow combines topic models and contextual embeddings to produce a scalar summary statistic of pairwise text similarity that spans multiple dimensions and can be aggregated over lawmaker-pairs, factions, parties, and time. After describing our workflow, we provide evidence of construct and convergent validity. Our measure aligns with human judgment of pairwise text similarity, and rhetorical cohesion correlates with several unidimensional measures of party talk (that do not correlate with each other). Then, we turn to two applications using the text of House floor speeches from the Congressional Record. First, we investigate four theoretically-grounded factors that should increase rhetorical cohesion: election timing, majority status, issue salience, and seat safety. Our measure generally recovers the expected positive relationships. Unidimensional measures, by contrast, can support, not support, or even reverse the expected correlations for theoretically sound reasons (given that each isolates a single dimension). For example, issue salience may plausibly increase or decrease a particular rhetorical dimension even if rhetorical cohesion is increasing in aggregate. Second, we apply our measure to a well-known historical pattern: we show that Southern and non-Southern Democrats exhibit stronger intra-factional rhetorical cohesion than cross-factional rhetorical cohesion before the 1990s, but these differences are much smaller after the 1990s. Our measure of rhetorical cohesion captures patterns that single-dimension proxies obscure, and it allows scholars to develop a deeper understanding of the rhetorical side of party branding.

Measuring Cohesion

A party's reputation is a collective good that benefits all members electorally (Aldrich 1995; Cox and McCubbins 1993, 2005; Lee 2009). A strong, consistent, and clear brand provides voters with heuristic information about what the party stands for, who it represents, and what it will do if it controls institutional power (Groeling 2010). For lawmakers, the brand provides a cheap way to signal complex information to voters. For voters, the brand provides information useful for both prospective and retrospective voting: by forecasting what the party will do in office and by providing a clear standard by which to hold it to account.

Valuable brands are fostered, in part, through consistency. When parties promote a clear and consistent message, they increase the likelihood that the public will receive, believe, and understand it (Groeling 2010). In particular, the American Political Science Association (1950) report on responsible parties highlights two key elements of this process: program articulation and program enactment. That is, voters must first know what the party commits itself to, and second, parties must act collectively to follow through on those commitments. These two elements map cleanly onto two distinct requirements: rhetorical cohesion (i.e., do co-partisans promote the same message) and legislative cohesion (i.e., do co-partisans vote as a bloc on these programs), which scholars have argued can independently strengthen a party's brand (on rhetoric specifically, see Groeling 2010; Hopkins 2018; Sellers 2009).

Yet despite general agreement that program articulation and program enactment are important elements of party brands, much of the research to this point has focused on roll-calls rather than rhetoric. Although voting behavior can tell us much about whether parties are voting cohesively, they tell us little about whether parties are promoting a unified message. We also have several text-based measures of party talk, however, these existing measures are generally unidimensional constructs that are purpose-built to test other theories about e.g., ideology, message diffusion, or homestyle. A general measure

of *rhetorical cohesion* requires that we measure party talk across a range of dimensions simultaneously—from topics and ideology to framing and justifications—to understand not only what partisans talk about, but how they present policy information to the public, and whether they do so in the same way. We note that this observable output can stem from a number of sources including leadership pressure and discipline, internal coordination, or coincidental alignment among members. Given that the literature has argued both parties and voters value rhetorical cohesion (Groeling 2010; Sellers 2009), our goal is to measure it reliably. We leave the sources of rhetorical cohesion to future work.

Roll-Call Measures Capture Enactment not Articulation

The roll-call record has long been the standard data source for understanding party cohesion. Whether direct applications like Rice scores and roll rates, or summary statistics like NOMINATE, these measures have provided important information about legislative cohesion and ideological positioning in Congress. However, these quantities speak directly to the program enactment side of the ledger. They tell us how often co-partisans vote together on policy, which is distinct from program articulation: the issues on which co-partisans speak and the ways in which they discuss them.

On a theoretical level, there is no requirement that what lawmakers vote on corresponds to the issues they emphasize on the campaign trail or on the floor. But even if it were, rhetorical and voting records are likely to diverge due to the dimensionality of the data. A bill is a bundled package designed by a coalition leader to attract sufficient support (Arnold 1990). Rank-and-file lawmakers face a binary choice between the bundled and the status quo. Setting aside issues of selection into the roll-call record (which also affect floor speeches), two paired alternatives can only tell us so much about party unity. Two co-partisans may vote in the same direction for different reasons (Duck-Mayr and Montgomery 2023) or vote together on a package where they would vote against one another on an unbundled set. Further, the trend toward omnibus bills and unorthodox

lawmaking further limit the rank-and-file's ability to defect on key votes (Curry and Lee 2020; Sinclair 2017).

Rhetoric, by contrast, is multidimensional. Two lawmakers can vote yes, but they can use rhetoric to explain which parts of a package they favor, what they oppose, or why they are voting yes. And when lawmakers differentiate in this way, they are reducing the party's rhetorical cohesion as compared to a case in which they both promote the logic for supporting a bill. Thus, voting cohesion does not fully capture message cohesion. For that, we need to look to what parties say about policy.

Existing Text Measures Don't Fully Capture Articulation

Scholars have long been interested in what lawmakers say, both from a theoretical perspective and methodologically, given the limited dimensionality of, and selection issues inherent in, the roll-call record. This interest has led to the development of several text-based measures of legislative rhetoric that have taught us valuable lessons about topic emphasis (Hughes and Koger 2022; Quinn et al. 2010), ideological positioning (Gaynor et al. 2025; Rheault and Cochrane 2020; Slapin and Proksch 2008), partisanship (Gentzkow, Shapiro and Taddy 2019; Peterson and Spirling 2018), nationalization (Ban and Kaslovsky 2024; Grimmer 2013; Noble 2024), and message diffusion (Gaynor 2025). Each of these measures can provide insight into rhetorical cohesion, but only partially.

To see why, consider the following excerpts from three speeches given by Republicans in opposition to the Affordable Care Act on November 3, 2009 in Table 1. First, note that all three Republicans voted against the bill when it came up for a vote four days later. But these lawmakers also justify their opposition on different grounds. The first two excerpts in Panel A, labeled Q1 and Q2, frame opposition in terms of the Medicare Advantage Program. Both representatives reference Democratic Majority Leader Pelosi, name the Medicare Advantage Program, and lament that millions of seniors will lose coverage. However, they use slightly different talking points: one referencing the "Pelosi health

Table 1: Illustrating Differences Across Text Similarity Measures

Panel A. ACA Quotes			
Q1	Rep. Joseph Pitts (R-PA): “The Pelosi health care bill we will consider later this week effectively eliminates the popular Medicare Advantage health plans that millions of seniors rely on for medical, vision, and dental care.”		
Q2	Rep. John Fleming (R-LA): “The CBO estimates in PelosiCare that it will cut over \$150 billion from Medicare Advantage...That will knock about 6 to 11 million seniors off of Medicare Advantage.”		
Q3	Rep. John Boozman (R-AR): “The newly created public option will be authorized to fund elective abortions. The Pelosi health care bill does not include the pro-life language...As the bill is written, Federal funds will pay for elective abortions.”		
Panel B. Similarity Measures			
Quote Pair	True Similarity	Topic Model	Text Reuse
Q1 & Q2	High	High	Low
Q1 & Q3	Low	High	Moderate
Q2 & Q3	Low	High	Low

care bill” and the other naming “PelosiCare.” The excerpt provided in Q3 differs more substantially. The speaker references “the Pelosi health care bill” as in Q1, but the justification revolves around abortion funding. It does not discuss the Medicare Advantage Plan or seniors’ healthcare coverage. These are real examples from the Congressional Record and reflect the fact the lawmakers can vote as a bloc while providing different framings, justifications, and talking points.

Intuitively, Q1 and Q2 have higher rhetorical cohesion to each other than either does to Q3, given their substantive and semantic overlap. However, whether we draw this conclusion depends on the measure we use to score text. A topic model would label all of these excerpts as highly similar, given that they each cover health care. A text-reuse method, like 5-gram overlap, would find few similarities between these three excerpts given limited textual overlap. In fact, it would code Q1 and Q3 as most similar given

their use of “the Pelosi health care bill” despite the different underlying justifications. Neither of these approaches are “wrong” in the sense that they each capture something real about variation in these messages: they all cover the same topics, and Q1 and Q3 use the same talking point. Rather, this disagreement reflects a deeper conceptual point that cohesion, as implied by theorists (APSA 1950; Fiorina 1980), is inherently multidimensional. It’s not (only) that parties use the same talking points or emphasize the same topics or express the same ideological positions; it’s that parties coordinate across all of these dimensions to present a coherent message about what they stand for and who they represent. A voter hearing Q1 and Q2 would likely infer that Republicans stand for seniors and dislike the ACA because it limits seniors’ healthcare access. A voter encountering Q1 and Q3 might be confused. Do Republicans care about cost? Do they care about seniors? Do they prioritize abortion? Even if a voter prefers one of these messages in isolation, in aggregate, these competing messages could dilute the party’s brand if voters process across dimensions simultaneously. This discussion highlights the need for a method of identifying rhetorical cohesion that can capture substantive alignment, aggregate across these dimensions, and summarize overall cohesion.

A General Framework of Measuring Rhetorical Cohesion

We draw on the legislative politics literature which defines party cohesion as “the extent to which, in a given situation, group members can be observed to work together for the group’s goal in one and the same way” (quoted in Hazan 2003, 3), in particular, as it relates to policy agreement (Sinclair 2003). We extend this idea to rhetoric, seeking to capture the extent to which ingroup members speak in a cohesive manner—such as expressing similar positions, ideologies, justifications, frames, etc—on a specific policy issue. This conceptual extension is consistent with recent work that applies party cohesion frameworks beyond formal legislative behavior, defining cohesion generally in terms of

semantic agreement in expressed messages.³

To help operationalize our definition, imagine a fictional “maximally cohesive party.” In a single-issue setting, we define such a party as one in which every member delivers an identical statement on the policy issue. Here, voters would be most likely to internalize the party’s message given the consistency and repetition (Groeling 2010). Of course, parties address many issues simultaneously, but maximal cohesion in our framework is issue-specific. We do not expect parties to articulate the same message on healthcare and civil rights policy, which are distinct issue domains. A maximally cohesive party in a multi-issue space is one that gives identical statements within issues. There may be some overlap—for example, if a party always invokes themes of limited government across policy areas—but it is not required.

Parties do not actually achieve this maximal level of cohesion. English is rich, and co-partisans express similar ideas using different words. Strategically, lawmakers’ messages deviate from their peers due to constituency concerns, ideological preferences, representational style, or poor leadership coordination. Maximal cohesion is generally infeasible and may not be desirable: co-partisans may appeal to different audiences with different messages. However, cohesion, as a construct, is necessarily about message similarity, and methodologically we seek to measure the degree of that similarity (which may decline for strategically sound reasons). We want a score that increases when a pair of documents approaches perfect overlap. The more any two documents differ—whether due to idiosyncratic differences or strategy—the lower our measure. By aggregating—over party members, over topics, and/or over time—we can produce a summary statistic of rhetorical cohesion at any unit of analysis.

We construct our measure in two stages. First, we identify documents that cover comparable policy issues using a supervised topic modeling framework. This step allows us to pair documents that discuss the same policy, which is essential for operationalizing

³See Mueller (2024), who explicitly adapts the definition of party cohesion to protest crowds, operationalizing the degree of agreement in the content and framing of participants’ expressed motivations.

our definition of rhetorical cohesion. Second, we capture message variation within issues by measuring the semantic similarity between co-partisans’ policy statements using contextual text embeddings. These two stages provide a fine-grained measure of rhetorical policy cohesion, i.e., speaking “in one way about the same issue.”

This approach differs from related work. First, our goals differ from scholars who measure partisanship and polarization (e.g., Gentzkow, Shapiro and Taddy 2019; Green et al. 2024; Peterson and Spirling 2018). Our interest is within-group similarities rather than between-group differences. Although these two quantities may correlate, they need not if a party contains distinct sub-factions that are still clearly separated from the opposition. Second, technological innovations have produced new ways to ideologically scale lawmakers using text (e.g., Gaynor et al. 2025; Porter and Case N.d.; Rheault and Cochrane 2020). While our measure of rhetorical cohesion is correlated with ideology (as shown below), it is conceptually distinct. Ideology is a single dimension, and ideologically dissimilar speeches may still be similar if they make use of emotional arguments, focus on national figures, etc.

Our measure is a scalar summary statistic of multidimensional rhetorical cohesion. As such, our measure imposes no left-right (or any other scaled) interpretation on the data. We know which pairs of speeches are more similar than others, but we cannot say, for any given pair, exactly which dimension(s) are driving the variation. As shown below, our measure correlates with unidimensional measures of textual similarity, and as we discuss in the conclusion, it is possible to decompose our measure and assess which unidimensional features drive the relationship (and even how those might change over time). Our intent is not to replace single-dimension text studies when they answer a theoretical question. Instead, we develop a measure specifically for theories that require a multidimensional summary of rhetorical cohesion across parties and time. We apply this general framework to U.S. House floor speeches, allowing us to construct measures of rhetorical cohesion among members of Congress.

Identifying Topics: Comparable Policy Issues Across Time

We demonstrate our approach with an application to U.S. House floor speeches. We rely on the post-Gingrich period, the 104th to 116th Congresses (1994–2020), to hold constant key features of the political environment, such as party polarization and electoral competition (Gentzkow, Shapiro and Taddy 2019; Lee 2016; Sinclair 2006). This choice mitigates against the possibility that observed variation in rhetorical cohesion is driven by changes in congressional procedure or the party coalitions. However, in our final application, we apply this same method to speeches given in the 87th to 103rd Congresses and examine how patterns of cohesion have changed over time. Although our two-stage workflow is agnostic to the corpus, floor speeches in the House are subject to many of the same agenda-setting and selection biases that affect roll-call voting. Given that roll-calls are the principle measure of program enactment, floor speeches are an appropriate comparison. Further, floor speeches ensure our estimates are comparable to other recent research exploring party cohesion (Gaynor et al. 2025; Gentzkow, Shapiro and Taddy 2019), and they are readily available and complete across our time series.

We begin by identifying the most prevalent topic of each speech using a keyword assisted topic model (keyATM, Eshima, Imai and Sasaki 2023). Unlike unsupervised LDA, keyATM allows researchers to nudge the topic model toward a set of topics using lists of pre-defined keywords. Our keywords come from the Comparative Agendas Project (CAP) master codebook (Jones et al. 2023), a project that categorizes political content into a set of 21 major topics such as macroeconomy, health, and foreign affairs. This model allows us to focus on a set of broad, stable policy issues that we expect to persist across our time series, facilitating over-time comparisons. To generate keywords, we downloaded Democratic and Republican Party Platforms that have been hand-coded at the quasi-sentence level by Wolbrecht et al. (2023). We calculated the tf-idf score of each stemmed word in this corpus within each topic and chose the top 15 words in each category as our topic-specific keywords (see Table A1 for the full list). We manually created

two additional non-CAP topics specific to congressional speeches: *parliamentary language* (e.g., quorum, yield) and *other*, which included uninformative and common words (e.g., people, think). We apply standard pre-processing to all speech transcripts (see Grimmer, Roberts and Stewart 2022, and Appendix A.2) and fit a separate keyATM model to all speeches delivered within each two-year Congress. This approach allows vocabulary and policy content to vary over time, while the use of keyATM stabilizes topics. After fitting, we assign each speech a single label according to its most prevalent topic and drop speeches labeled *parliamentary* or *other*, given our focus on *policy* cohesion.⁴ Following Ying, Montgomery and Stewart (2022), we validate that speeches are substantively related to their assigned topics in Appendix A.3.

Beyond Topics: Capturing Party Cohesion with Semantic Similarity

Topics are inherently coarse and do not capture the nuances of how lawmakers frame, justify, or take positions on policy issues. Moreover, floor speech topics are shaped by procedural rules and party leaders’ agenda-setting power. We therefore move beyond topic-level analysis to measure the substance of political communication using state-of-the-art text embeddings. Specifically, we employ OpenAI’s commercial contextual embedding model (text-embedding-3-small, hereafter OpenAI-3S), alongside BGEM3, an open-source embedding model trained for semantic retrieval using both dense and ColBERT representations.⁵

These embedding models offer three advantages. First, both OpenAI and BGEM3 em-

⁴Although speeches are characterized by mixed-membership topic distributions, incorporating this information would require computing weighted embedding similarities across all pairs of speeches and topics in the next step. Given the scale of congressional speech data, doing so is computationally expensive. Moreover, our definition of party cohesion is issue-specific. Accordingly, we assign each speech to its highest-probability topic and conduct within-topic comparisons.

⁵BGEM3 is a transformer-based retrieval model that produces both dense embeddings, which represent documents as single vectors optimized for semantic similarity, and ColBERT-style embeddings, which encode texts as sets of contextualized token-level vectors enabling fine-grained, interaction-based similarity matching (Chen et al. 2024). In contrast, OpenAI-3S models are derived from LLMs in the GPT family and learn contextual semantic representations from large-scale corpora using language-modeling-based objectives. Full details of their architecture and training data are not publicly disclosed (Yenduri et al. 2024).

beddings leverage transformer architectures that learn word meanings in context through language-modeling or contrastive retrieval objectives. As a result, texts that express similar ideas, even when using different words, are represented as “close” in the embedding space. These models capture semantic relatedness rather than exact word overlap. Second, these models produce fully contextual embeddings; token representations vary with their usage, which is important in political contexts. They improve upon static word embeddings such as word2vec (Mikolov et al. 2013) and GloVe (Pennington and Manning 2014), which assign fixed vectors to words regardless of context (see Lin 2025, for a comprehensive discussion). Third, advances in model scale and architecture (Minaee et al. 2024) allow these embedding models to handle substantially longer input sequences than earlier pre-trained language models. In particular, they support inputs of up to 8,191 tokens, enabling us to encode nearly every speech in our corpus without truncation.⁶ Models like BERT (Devlin et al. 2018) are limited to 512 tokens and typically require task-specific fine-tuning to achieve strong performance (Wang 2023). Existing research also demonstrates that LLMs and their derived embeddings generalize well in zero- and few-shot settings (Rathje et al. 2024; Tao et al. 2024), whereas modern retrieval-trained embeddings such as BGEM3 provide strong off-the-shelf performance for semantic similarity, making both suitable for large, unlabeled political text corpora.

To compute our measure, we pair speeches at the day-party-topic level. That is, we pair each speech given by a lawmaker on a topic with each other speech given by a co-partisan member on the same topic on the same day.⁷ By restricting pairs to the same period of time, we maximize the likelihood that speeches on the same topic cover the same underlying sub-issue. However, researchers can choose their unit of analysis as befits their study. For example, a scholar interested in party factions could aggregate at

⁶We drop the vanishingly small number of speeches that exceed this maximum length.

⁷We exclude independents and party-switchers in the Congress they switch. Throughout, all covariates come from VoteView (<https://voteview.com/>) and The Center for Effective Lawmaking (<https://thelawmakers.org/>). We exclude a small subset of lawmakers for whom we do not have key covariates.

the caucus level; someone interested in differences between foreign and domestic policy could aggregate across coarser topics.

Once paired, we output a scalar similarity score that captures the degree of rhetorical cohesion between the two texts. For OpenAI-3S and BGEM3-dense models, we use cosine similarity scores (as recommended in the documentation), which theoretically range from -1 to 1 , with higher values indicating greater similarity between statements and negative values indicating dissimilar or unrelated inputs. In practice, we do not observe negative scores as speeches are pre-paired within the same topic and party. The BGEM3-ColBERT model aggregates token-level cosine similarities using a late-interaction mechanism, matching each token in one statement with its most similar token in the other and averaging the resulting similarities to produce a normalized semantic alignment score between 0 and 1 .

Validating our Measures

In this section, we assess the construct and convergent validity of the embedding-based similarity scores.

Construct Validity Against Human Judgment

To determine whether our similarity measure derived from contextual embeddings accurately captures rhetorical cohesion, we worked with a trained RA to conduct a novel pairwise validation exercise (cf. Carlson and Montgomery 2017). We presented the RA with a *focal speech* sampled from our corpus along with two *comparison speeches* that were paired with the focal speech in the underlying data.⁸ The RA was asked to complete a

⁸We worked with a single trained RA given the demanding and complex nature of the task. Speeches in our sample are long, and these comparisons require careful reading, sustained attention, and thoughtful judgment over a span of weeks. Given cost and time constraints, our RA was able to complete 250 valid comparison tasks.

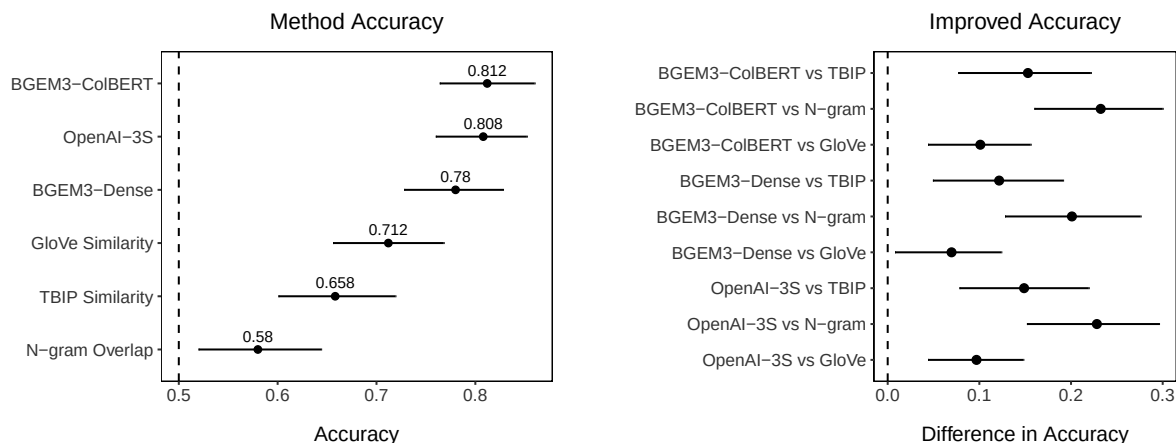
series of *comparison tasks*. For each, he carefully read the focal speech and both comparison speeches, then he selected the comparison speech that was more similar to the focal speech, where similarity was broadly defined. We sampled several focal speeches, and for each focal speech, we drew several comparison speeches paired to that focal speech. This ensured we were able to validate the accuracy both within a given focal speech and then across several sets of focal-comparison sets. Additional detail on this validation task is in Appendix A.4.

Our quantity of interest is the proportion of times a given measure selects the comparison speech chosen by our RA. In total, the RA and similarity scores derived from three contextual embedding models—OpenAI-3S, BGEM3-ColBERT, and BGEM3-Dense—matched in 80.8%, 81.2%, and 78% tasks, respectively, which is strong given the complexity. As a further demonstration, we compare the performance of our measure to related single-dimensional measures. These alternatives include the negative absolute difference of a pair’s text-based ideal point (TBIP) scores (Gaynor et al. 2025),⁹ 5-gram text overlap (Casas, Denny and Wilkerson 2020; Gaynor 2025), and the cosine similarity of the pairs’ averaged GloVe vectors from the Stanford pre-trained embeddings (Rodriguez and Spiraling 2022). We code a match as when, for the RA selected focal-comparison pair, the absolute difference is smaller (TBIP), the set Jaccard similarity is higher (5-gram overlap), the cosine similarity score is higher (GloVe).

Figure 1 shows that our three measures of rhetorical cohesion best recover the selections made by our trained RA. In the left panel, their bootstrapped accuracy ranges from 0.780 to 0.812, with the BGEM3-ColBERT score performing best. GloVe achieves a significantly lower accuracy of 0.712 than the three contextual embedding models at the 95% level, as presented in the right panel, where statistical differences are assessed via bootstrapping. TBIP similarity and N-gram overlap, which capture specific dimensions of

⁹Given the structure of our data, we construct Congress-topic pairs from our existing data and fit a Wordfish model on each of these subsets in our full dataset rather than computing a true TBIP score. We exclude speeches for which the model did not converge.

Figure 1: Accuracy of Rhetorical Cohesion and Single-Dimension Benchmarks in Recovering Human Coding



Note: The left panel depicts accuracy of our rhetorical cohesion measure and the alternatives in recovering the human-selected more-similar speech. Our rhetorical cohesion measures matched our RA’s selection at higher rates than other text measures of related concepts. The 95% confidence intervals are generated via 1,000 bootstraps. The right panel depicts the pairwise accuracy differences with bootstrapped confidence interval. The accuracies of our methods are statistically distinguishable from those of alternative approaches we tested.

text, perform substantially worse, with accuracies of 0.658 and 0.580, respectively. Over 1000 bootstraps, we demonstrate the embedding-based methods are significantly more accurate than the single-dimension measures at recovering human judgments of text similarity.

Convergent Validity: Correlations with Existing Proxies

We have shown that our measure of rhetorical cohesion accords with human judgments of textual similarity. However, our previous results do not provide insight into what our measure is capturing. Our RA’s selections may be driven by a single measure, such as ideological emphasis or shared phrases, rather than a more multidimensional understanding of similarity. If our measure is truly multidimensional, it should positively correlate with unidimensional proxies of textual similarity, but it should not be fully explained by any single one. To test this, we compute the Pearson correlation between

Table 2: Correlation Between Rhetorical Cohesion and Relevant Proxies

	OpenAI	Dense Dot	ColBERT	NOMINATE	Both Pres. Ref.	Text Overlap	TBIP
OpenAI	—						
Dense Dot	0.889	—					
ColBERT	0.864	0.901	—				
NOMINATE	0.054	0.056	0.048	—			
Both Pres. Ref.	0.174	0.148	0.156	0.020	—		
Text Overlap	0.128	0.162	0.142	0.020	-0.022	—	
TBIP	0.402	0.375	0.344	0.031	0.049	0.055	—

Note: All pairwise correlations are statistically significant at $p < 0.001$.

our measure and existing proxies for pairwise similarity from the literature: NOMINATE scores, TBIP (Gaynor et al. 2025), presidential references (Groeling 2010), and 5-gram text overlap (Gaynor 2025; Casas, Denny and Wilkerson 2020).

For our quantities of interest, we compute the negative absolute difference of the speakers’ NOMINATE scores, the negative absolute difference of TBIP estimates of paired speeches, a binary indicator for whether both speeches reference the president,¹⁰ and the set Jaccard similarity of 5-grams in both speeches, respectively. We expect each of these metrics to be positively correlated with our measure of rhetorical cohesion, however, we expect these correlations to be modest or low given that they are four of many sub-components that comprise rhetorical cohesion.

We present pairwise correlations in Table 2. All three focal measures appear in the upper-left panel of the table and are highly correlated, with coefficients ranging from 0.864 to 0.901. In the bottom left of the table, we can see how our focal measures of rhetorical cohesion correlate with existing single dimension measures of legislative and rhetorical behavior. First, we see that NOMINATE similarity has a positive, but small, correlation with rhetorical cohesion between 0.048 and 0.056. Members who have more ideologically similar voting records give more similar speeches, but the relationship is weak.

¹⁰We follow Noble (2024) in coding a reference as when the president’s last name or “the president” is used.

The low correlation between program enactment and program articulation further underscores the importance of studying and measuring rhetorical cohesion as a standalone concept in the party unity literature. Next, we turn to text-based measures of similarity and find stronger, but still weak to moderate, correlations. Presidential references are correlated at 0.159 on average, and text overlap is correlated at 0.144 on average. The latter is particularly interesting given that, mechanically, our measure of rhetorical cohesion will increase the more overlapping 5-grams are present in a speech. To the extent that two speeches are similar, that similarity is being driven more by meaning than shared words or phrases. Finally, TBIP shows the highest correlation with our measure, 0.374 on average. This correlation is modest, but note that it explains only about 14% of the variance ($R^2 = 0.14$) of our measure. They are related, but our measure is distinct from topic-based and ideology-based measures of cohesion. The pairwise correlations between alternative measures, in the bottom right of the table, are also of interest. Although each of these measures is positively correlated with ours, as expected, the correlations between these alternatives tend to be much weaker—and in one case, negative.

The coefficients in Table 2 yield three important insights about our measure. First, these three different embedding models produce similar substantive outputs. As we show later, results using any of these measures are consistent. Second, our measure of rhetorical cohesion captures similarity in a multi-dimensional sense. It is positively correlated with standard legislative and text measures that have been used in the existing literature to draw conclusions about lawmaker similarity. This is exactly what we would expect to see if our measure were absorbing single-dimension measures of textual similarity. Third, the existing unidimensional measures do not strongly correlate with one another. Although there are good reasons to use a single-dimension measure for a theoretically motivated study, the results in the table highlight the risk of using any one of these as a standalone proxy for rhetorical cohesion. Taken together, this analysis shows that our measure captures a multidimensional understanding of rhetorical cohesion that

cannot be easily explained by any single dimension.

Applications

Having validated our measure of rhetorical cohesion, next, we demonstrate how our measure can be used to uncover new insights and directions to explore in the congressional politics literature.

Explaining Variation in Rhetorical Cohesion

The party brand, produced through cohesive behavior, is a collective good that benefits all members. Yet as with all collective goods, some individuals face incentives to defect and free ride off the efforts of others. These incentives may vary with the political environment, leading to periods of higher or lower party cohesion. However, existing tests of these theories either rely on measures of roll-call cohesion or single dimensions of text. Our multidimensional measure of rhetorical cohesion, validated against human judgment, provides new leverage to test these theories in legislative speech. Here, we focus on the post-1994 period, where increasing congressional polarization should heighten the incentives for rhetorical cohesion (Gentzkow, Shapiro and Taddy 2019; Noble and Butler 2026; Sinclair 2017). We examine four theoretically grounded correlates of cohesion: electoral proximity, majority control, issue salience, and seat safety. As we show, our measure generally captures the expected positive relationships, whereas plausible single dimension proxies can sometimes produce inconsistent results (given that each dimension need not move in the same direction).

To the extent that party brands are valuable commodities, normative and positive theorists agree that they are most relevant in the electoral context. A consistent message allows voters to hold politicians accountable (Aldrich 1995; APSA 1950), and a strong party brand improves a party's electoral performance (Cox and McCubbins 2005; Lee

2009). Although individuals may face incentives to defect, a logical implication of these theories is that pressure for cohesion should increase as elections approach. Consistent with this interpretation, Weber and Parsons (2016) find that roll-call cohesion increases as election day nears. We expect to find a similar positive correlation between the electoral cycle and rhetorical cohesion.

The primary prediction of Cartel Theory (Cox and McCubbins 1993, 2005) is that the majority party will control the legislative agenda to promote unity within the caucus. By gatekeeping divisive bills, the majority only votes on issues over which they agree, and ideally, on issues that split the minority. Here, we focus on House floor speeches. In this setting, speech must be germane to the legislative agenda—and that legislative agenda is endogenously chosen by the majority. As a consequence, we expect higher rhetorical cohesion among the majority than the minority given the institutional constraints of this setting.

We argue that cohesion is issue specific, however, not all issues demand the same level of cohesion. Lawmakers should have stronger incentives to engage in cohesive behavior over issues that are more salient. These issues should receive more attention from the media and voters, and therefore, will be those that most impact the party brand—raising the cost of defection (Groeling 2010; Sellers 2009). In a survey of Danish politicians, Tromborg (2019) finds this pattern: rank-and-file positions are closer to leaders' positions when issues are more salient. We expect to find a similar relationship in rhetoric.

Finally, we expect that some lawmakers will be more likely to behave cohesively than others. In particular, lawmakers in marginal seats face tradeoffs between echoing a party message and catering to their own constituencies. These incumbents may be punished by constituents for excessive party loyalty in voting behavior, whereas those in safer seats may be rewarded for it (Canes-Wrone, Brady and Cogan 2002; Carson et al. 2010). Rhetorically, marginal members are likely to de-prioritize national issues and symbols in their constituency communications (Grimmer 2013; Noble 2024). Ultimately, we expect safe-

seat lawmaker-pairs to evince more rhetorical cohesion than other dyadic combinations.

To test these hypotheses, we regress our three measures of rhetorical cohesion (OpenAI-3S scores, BGEM3-Dense dot, and BGEM3-ColBERT scores) on a set of key independent variables. Our unit of analysis is a speech-dyad (which is clustered within lawmaker-dyads, topics, and congresses). We interpret factors that systematically increase or decrease speech-dyad similarity as, more broadly, reflecting changes in rhetorical cohesion across the party. To measure electoral proximity, we follow Noble (2025) and code speeches given on all days between Labor Day and the November election in even years as occurring during a *Midterm Election Period* or *Presidential Election Period*. We separate these two types of elections given the divergent electorates and potentially different messaging incentives lawmakers face in these two distinct environments. For majority, we simply code a dyad as 1 if their party controls the House majority or 0 otherwise. To assess issue salience, we leverage the Comparative Agendas Project version of the Gallup Most Important Problem question. This data is aggregated to the year-level and reports the proportion of respondents naming a problem as the most important in that year. These data are organized into the same 21 CAP categories from our topic models and can therefore be matched. Finally, we measure seat safety as a trichotomous variable. We code a speech-dyad as “both safe” if both speakers represent a district in which their party’s presidential candidate won at least 65% of the vote (in line with Carson et al. 2010). The two other categories are (i) dyads where neither member represents a district with greater than 65% vote share for the party’s presidential candidate and (ii) a mixed pair with one greater and one below 65% presidential support. Our models additionally control for whether the lawmakers are presidential co-partisans, the number of House leaders in the pair (from Volden and Wiseman 2014), the pair’s party, the NOMINATE similarity between two members (i.e., negative of NOMINATE distance), and whether two lawmakers represent the same state. We also include topic and Congress fixed effects and present models with and without lawmaker-dyad fixed effects. Standard errors are clustered at

Table 3: Correlates of Rhetorical Cohesion

	OpenAI-3S		BGEM3-Dense		BGEM3-ColBERT	
	(1)	(2)	(3)	(4)	(5)	(6)
Pre-Midterm Election	0.024*** (0.001)	0.023*** (0.001)	0.009*** (0.001)	0.009*** (0.001)	0.012*** (0.001)	0.012*** (0.001)
Pre-Presidential Election	-0.021*** (0.001)	-0.015*** (0.001)	-0.017*** (0.001)	-0.013*** (0.001)	-0.015*** (0.001)	-0.011*** (0.001)
Majority	0.005*** (0.001)	0.003*** (0.001)	0.006*** (0.000)	0.005*** (0.000)	0.005*** (0.000)	0.004*** (0.000)
Issue Salience (Pct. Most Important)	0.046*** (0.003)	0.042*** (0.002)	0.023*** (0.002)	0.023*** (0.002)	0.008*** (0.002)	0.007*** (0.002)
Both Lawmakers in Safe Seat	0.005*** (0.001)	0.000 (0.001)	0.003*** (0.001)	-0.000 (0.001)	0.002** (0.001)	-0.001 (0.001)
One Lawmaker in Safe Seat	-0.000 (0.001)	-0.001 (0.001)	-0.000 (0.000)	-0.001+ (0.001)	-0.000 (0.000)	-0.001 (0.001)
Republican	-0.023*** (0.001)		-0.016*** (0.000)		-0.017*** (0.000)	
NOMINATE Similarity	0.038*** (0.002)		0.027*** (0.001)		0.020*** (0.001)	
Presidential Co-Partisans	-0.001** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	-0.002*** (0.000)	0.000 (0.000)	-0.000 (0.000)
N. Leaders in Pair	0.016*** (0.001)	0.007*** (0.001)	0.010*** (0.000)	0.004*** (0.001)	0.010*** (0.000)	0.004*** (0.001)
Same State	0.013*** (0.001)		0.009*** (0.001)		0.006*** (0.001)	
Fixed Effects						
Lawmaker Dyad		✓		✓		✓
Congress	✓	✓	✓	✓	✓	✓
Topic	✓	✓	✓	✓	✓	✓
Num.Obs.	1635522	1604737	1635522	1604737	1635522	1604737
R2 Adj.	0.062	0.186	0.069	0.190	0.057	0.179
R2 Within Adj.	0.014	0.002	0.013	0.002	0.011	0.001

Note: Results are from ordinary least squares models. Standard errors are clustered at the lawmaker-dyad level.

the dyad level.

We report results in Table 3 across two model specifications for three different versions of our dependent variable.¹¹ In the first row, we show that between Labor Day and Election Day of a midterm year, lawmaker dyads become more rhetorically cohesive across all model specifications. The magnitude of the effect ranges from 0.009 to 0.024 on the unit scale, or about 0.13 of a standard deviation shift on average. In line with

¹¹The dependent variables are reported on their original scales. Although they have different theoretical ranges, their observed ranges are comparable. Our goal is to demonstrate consistency in sign, significance, and substantive pattern rather than compare coefficient sizes across models.

the theoretical prediction, when midterm elections approach, parties increase rhetorical cohesion. However, as we show in the second row, parties become *less* rhetorically cohesive as presidential elections approach. We did not anticipate this finding, however, the effect is consistently signed and significant across specifications (and below, across unidimensional measures), suggesting it is not a measurement artifact, but rather, a consistent finding that deserves theoretical investigation in future work. We speculate that presidential campaigns complicate congressional messaging. During midterms, congressional parties have more control over the policies and issues they wish to discuss. The president's party may tout their policy wins over the past two years while the out-party likely aligns their messaging in opposition to the sitting president (cf. Lee 2016). During presidential elections, both sides have to choose how to respond to campaign themes promoted by the dueling presidential campaigns. A presidential nominee's message may not work well for every member's unique constituency, requiring that they adapt or distance themselves, decreasing rhetorical cohesion.

Next, we turn to the effects of majority control in the third row. Here, we see that majority parties are about 0.005 points (about 0.05 standard deviations) more rhetorically cohesive than minority parties, consistent with predictions stemming from Cartel Theory (Cox and McCubbins 1993). In row four, we support our prediction that issue salience should positively correlate with rhetorical cohesion. Finally, rows five and six provide insight into how seat safety and cohesion correlate. The omitted category is when neither lawmaker is safe, and we can see that when both lawmakers are safe, by comparison, cohesion is about .003 units higher. However, this result is cross-sectional; the null results in the lawmaker-dyad models suggest that lawmakers do not adapt their behavior as their district changes (Bernhard and Sulkin 2018, cf.), but rather, these differences are driven by fixed member characteristics. Ultimately, the raw effect size due to each covariate and the R^2 is modest, which makes sense given the number of factors that can affect speech similarity and the high dimensionality of text data. Importantly, the sign, size, and

Table 4: Correlates of Rhetorical Cohesion Vary Across Unidimensional Measures

	BGEM3-Colbert	TBIP	Pres. Ref.	Text Reuse
	(1)	(2)	(3)	(4)
Pre-Midterm Election	0.121*** (0.009)	0.085*** (0.009)	0.221*** (0.007)	0.094*** (0.010)
Pre-Presidential Election	-0.128*** (0.011)	-0.101*** (0.008)	-0.026*** (0.007)	-0.020* (0.009)
Majority	0.057*** (0.004)	-0.008* (0.004)	0.132*** (0.003)	-0.042*** (0.002)
Issue Salience (Pct. Most Important)	0.197*** (0.023)	-0.221*** (0.021)	1.255*** (0.020)	-0.436*** (0.012)
Both Lawmakers in Safe Seat	0.014* (0.006)	0.029*** (0.004)	0.031*** (0.004)	0.018** (0.006)
One Lawmaker in Safe Seat	-0.006 (0.004)	0.006+ (0.003)	0.018*** (0.003)	-0.003 (0.003)
Republican	-0.198*** (0.005)	-0.062*** (0.004)	-0.088*** (0.003)	-0.054*** (0.002)
NOMINATE Similarity	0.225*** (0.016)	0.204*** (0.013)	0.146*** (0.010)	0.068*** (0.011)
Presidential Co-Partisans	0.004 (0.003)	-0.015*** (0.003)	-0.208*** (0.002)	0.073*** (0.002)
N. Leaders in Pair	0.106*** (0.005)	0.048*** (0.003)	0.044*** (0.004)	0.029*** (0.005)
Same State	0.069*** (0.008)	0.038*** (0.005)	-0.025*** (0.005)	0.055*** (0.008)
Fixed Effects				
Congress	✓	✓	✓	✓
Topic	✓	✓	✓	✓
Num.Obs.	1510023	1510023	1510023	1510023
R2 Adj.	0.056	0.037	0.195	0.099
R2 Within Adj.	0.013	0.004	0.025	0.006

Note: Results are from ordinary least squares models. Standard errors are clustered at the lawmaker-dyad level. We subset to complete cases across all four dependent variables.

significance is consistent across specifications, indicating that these factors do robustly correlate with rhetorical cohesion as expected.

In Table 4, we present models using a slate of unidimensional measures of text similarity. Model 1 is simply a reproduction of model 5 in Table 3 for comparison. Model 2 uses our measure of text-based ideal point scaling. The dependent variable is negative

absolute value of the speech-pair Wordfish scores within-topic. Model 3 uses a binary indicator of whether both speeches in the pair reference the sitting president at least once. The last column is the set Jaccard similarity of 5-grams in both speeches. Positive coefficients indicate that a given factor correlates with an increase in rhetorical similarity across that given dimension. We standardize all four dependent variables to facilitate cross-model comparisons.

Comparing across models, we observe a number of differences as well as similarities. First, irrespective of whether we use our multidimensional measure or a unidimensional proxy, we see that parties rhetorically cohere when midterms approach and when we compare members representing safe seats to those who do not. That all dependent variables exhibit consistent patterns highlights the convergent validity of our measure and points toward two robust findings about the relationships between these variables and rhetorical cohesion. However, we observe a number of significant differences across other covariates. Although issue salience and majority control are associated with expected increases in our human-validated measure of rhetorical cohesion, these conclusions vary across unidimensional measures. It appears that issue salience is associated with reduced ideological similarity and direct text reuse among lawmakers. Yet, this factor is associated with a large increase in presidential references. We also see that majority dyads are *less* ideologically similar than minority dyads and majority dyads are also less likely to directly reuse co-partisan rhetoric. But majority parties are much more likely to talk about the president. We also observe differences in how lawmakers leverage presidential references when only one lawmaker in a dyad is in a safe seat. Although Cartel Theory predicts increased cohesion among the majority, this seems to be driven by an increase in partisan rhetoric rather than an increase in ideological similarity. As in the previous section, we highlight the fact that not only do these results differ, at times, from our human validated measure, but they disagree with one another, underscoring the need for a multidimensional measure of cohesion.

Our intent in highlighting these differences is *not* to say that any given single-dimensional measure is misleading or wrong. Rather, we use Table 4 to demonstrate that rhetorical cohesion is a multidimensional construct, and single dimension summaries provide a different kind of information rather than serving as a more straightforward proxy. Rhetorical cohesion can increase on net even when single dimensions decrease for theoretically valid reasons. For example, we speculate that salient issues could induce less ideological extremism among lawmakers as they seek to appeal to the widest possible audience, but rhetorical cohesion could increase overall as lawmakers speak in less technical and more easily interpretable vocabulary. We advise scholars to choose their measure wisely. Those interested in a general measure of rhetorical cohesion should follow our workflow given that it is human validated and recovers these theoretically expected relationships. Scholars with specific theoretical predictions about a single text dimension should continue to use the existing suite of measures.

Southern Democrats

In the previous section, we demonstrated that our measure of rhetorical cohesion shows expected correlations with variables that have been theorized but rarely tested in the existing literature. In this section, we apply our measure to a well-documented historical case to provide evidence of validity beyond our human-validated corpus. To do so, we apply the same methodology to estimate rhetorical cohesion in House floor speeches back to the 86th (1961) Congress.

Throughout the twentieth century, the Democratic party housed a distinct Southern faction whose ideological preferences diverged from the non-Southern faction of the party. In the mid-twentieth century, they often voted together at higher rates than other groups in the House, including non-Southern Democrats and Republicans (Key 1949), and they routinely allied with the Republican party against non-Southern Democrats when issues of civil rights were at stake. Beginning with the Civil Rights movement in

the 1960s, and accelerating with the Southern Realignment in the 1980s and early 1990s, conservative Southern Democrats were replaced by conservative Southern Republicans, polarizing the two parties (McCarty, Poole and Rosenthal 2008; Sinclair 2006).

Given patterns in roll call voting, we expect that a valid measure of rhetorical cohesion would recover inter-factional differences between Southern and non-Southern Democrats. In particular, we hypothesize that same-faction pairs (i.e., two Southern Democrats or two non-Southern Democrats) will exhibit higher rhetorical cohesion as compared to a cross-factional pair (i.e., one Southern and one non-Southern Democrat) before the end of the Southern Realignment. We define our cut-point for this analysis in 1995, the Republican takeover of the House, when many Southern Democrats lost their seats and which several scholars describe as a key turning point in party politics and polarization (Lee 2016; Sinclair 2006; Theriault and Rohde 2011). After this period, we expect these differences to substantively shrink between factions as conservative Southern Democrats lost elections and those who remained became more liberal.

To test our hypothesis, we apply our workflow to same-party pairs of House floor speeches delivered between the 87th–116th Congresses. Our final dataset comprises over 4.6M speech-pairs. We use the Colbert score as our dependent variable given its superior performance in the human-validation task. First, we separate pairs into four distinct groups: non-Southern Democratic pairs (the baseline group), Southern Democratic pairs,¹² cross-factional Democratic pairs, and Republican pairs. We then regress the Colbert Score on this indicator variable which we interact with a dummy for pre/post-1994. Our models include topic and Congress fixed effects, and we cluster standard errors at the pair-level.

We present results in the form of a marginal effects plot in Figure 2 (Table B1 in Appendix B.1 contains the full regression model). To interpret the plot, note that all coefficients are relative to the baseline category: a pair of non-Southern Democratic lawmakers,

¹²These include Democrats from states that seceded during the Civil War: AL, AR, FL, GA, LA, MS, NC, SC, TN, TX, and VA.

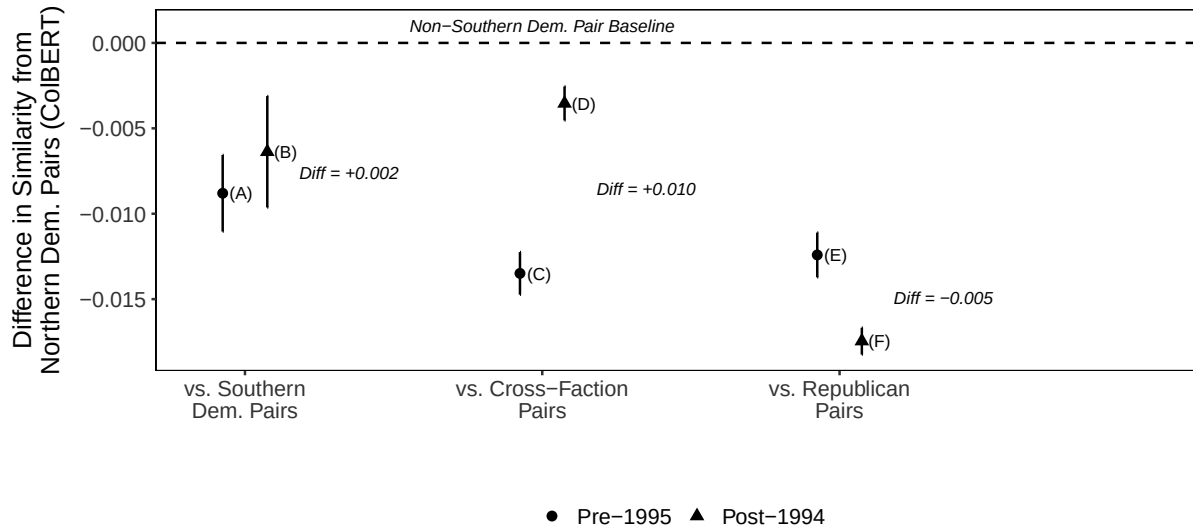


Figure 2: Rhetorical party cohesion of Southern Democratic pairs, cross-faction (i.e., one Southern and one non-Southern) pairs, and Republican pairs as compared to non-Southern Democratic pairs. Diff values indicate the change between the pre- and post-period for each pair-type. Before 1995, Democratic cross-factional pairs exhibit low rhetorical cohesion, but this quantity increases substantially after 1995, when many conservative Southern Democrats lost their seats and the Democratic party became more internally homogeneous.

whose rhetorical cohesion is normalized to zero. All coefficients in the plot fall below the dashed line, indicating that all other pair types are *less* rhetorically cohesive than two non-Southern Democrats in both periods. Circles represent pre-1995 rhetorical cohesion, while triangles represent post-1994 cohesion.

Beginning on the left, we see that two Southern Democrats in the pre-1994 period are about 0.009 points less cohesive than two non-Southern Democrats in this period (coefficient A). However, cross-factional pairs are even less rhetorically cohesive: -0.013 (coefficient C). As expected, cross-factional Democratic pairs are less cohesive than either same-faction Democratic pair-type. However, these patterns change after 1994. Although Southern Democratic pairs become modestly more cohesive (coefficient B vs A), cross-factional pairs become significantly more cohesive: an increase of 0.010 points (co-

efficient D vs C). Although non-Southern Democratic pairs still exhibit the highest level of rhetorical cohesion in the post-1994 period, there is little difference between Southern Democratic pairs and cross-factional pairs. This result is consistent with our hypothesis: after the Gingrich Revolution and the decline of the Southern Democrats as a distinct voting bloc, rhetorical distinctions between Southern and non-Southern Democrats attenuate significantly. Although we did not have a specific prediction for Republicans, we see that the party evinces low rhetorical cohesion across both periods as compared to non-Southern Democratic pairs.¹³ Taken together, these results recover the expected factional and temporal patterns and demonstrate the face validity of our measure on a historical corpus.

Conclusion

Understanding how political parties coordinate their public messaging is central to the study of legislative behavior, political communication, and democratic accountability. While scholars have developed a rich theoretical literature on party unity, empirical work has focused primarily on legislative cohesion, particularly roll-call voting. Much less attention has been paid to the programmatic articulation of party unity—whether politicians emphasize the same ideas when discussing policy issues. To address this challenge, we introduce a scalable measure of rhetorical cohesion that combines topic modeling with contextual embeddings to estimate the semantic similarity of speeches within policy domains. This approach allows researchers to measure whether members of the same party emphasize similar ideas when discussing the same issue areas, thereby providing a multidimensional indicator of party unity in political messaging.

Across our validation exercises and empirical applications, the resulting measure be-

¹³Note, because all coefficients are relative to the non-Southern Democratic baseline, the decline among Republican pairs (coefficient F vs E) is driven by a large increase among the baseline group and a smaller increase among Republicans—not a decline in Republican rhetorical cohesion.

haves in ways that are consistent with theoretical expectations. Rhetorical cohesion varies systematically with factors such as electoral proximity, majority status, issue salience, and seat safety. We also find that rhetorical cohesion declines around presidential elections, suggesting that parties temporarily loosen message discipline during periods of intense electoral competition, which merits further research. At the same time, single-dimension text measures for rhetorical similarity, such as ideological scaling, presidential references, or keyword overlap, produce inconsistent results across these same tests. This does not indicate that existing measures are flawed; rather, it highlights that rhetorical cohesion is inherently multidimensional and cannot be adequately captured by any of these textual proxies alone.

A central component of our framework is the use of contextual embedding models to estimate semantic similarity between speeches. In our implementation, we show that several state-of-the-art models—both commercial and open-source—perform comparably well in recovering human judgments of rhetorical similarity. Specifically, OpenAI-3S and two configurations of the BGEM3 model (dense and ColBERT) all achieve high levels of accuracy. Despite differences in architecture and training data, the resulting estimates of rhetorical cohesion are highly correlated. This convergence suggests that modern contextual embedding models have reached a level of maturity that allows researchers to obtain reliable measures of semantic similarity using either proprietary APIs or freely available open-source packages.

More broadly, the contribution of this paper lies not in any particular embedding model but in the general workflow we propose for measuring rhetorical cohesion. Topic models are first used to identify the issue spaces in which politicians communicate, ensuring that speeches are compared within substantively meaningful policy domains. Contextual embeddings then estimate semantic similarity among speeches within each issue space, capturing both shared vocabulary and deeper relationships in meaning. This combination allows scholars to estimate rhetorical cohesion while preserving the multidimen-

sional nature of political messaging.

Like any text similarity method, our approach has limitations. One challenge concerns the computational cost of pairwise comparisons. Measuring similarity among n documents requires $O(n^2)$ operations, which becomes expensive for large corpora. However, we can mitigate this challenge through blocking strategies. For example, researchers may restrict comparisons to speeches addressing the same policy issue or legislative agenda category within the same time window. Such theoretically informed blocking rules substantially reduce the number of comparisons required while ensuring that the resulting similarity estimates remain substantively meaningful.

Although we focus on party cohesion in the U.S. Congress, the framework can be easily adapted to other contexts. Scholars may apply the same workflow to comparative legislative settings and measure rhetorical cohesion globally. If they adopt a consistent topic classification scheme—such as the Comparative Agendas Project coding framework—the resulting similarity metric could facilitate comparisons of rhetorical cohesion across countries and issues on a common scale. Our workflow is also useful for studying policy diffusion and legislative hitchhiking—allowing researchers to understand in-spirit versus verbatim adoption.

The ability to measure rhetorical cohesion opens a number of new avenues for research. Future work could extend our analysis of the factors that drive rhetorical cohesion. It may also be fruitful to investigate the stylistic dimensions of cohesion—for example, whether emotional appeals, partisan language, or moral rhetoric influence the degree to which party messaging converges. By providing a scalable method for measuring rhetorical cohesion, the framework developed here allows scholars to study not only how parties vote together, but also how they speak with one voice in the political arena.

Disclosures. We used models from OpenAI and Anthropic to assist with writing code and project management as well as to provide feedback on drafts. The authors take responsibility for the content and accuracy of all content shared in this manuscript.

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Rhetorical Cohesion

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A Measurement of Message Discipline

A.1 Keywords for keyATM Model

In Table A1, we present the set of topics and keywords used to fit our keyATM models. These keywords are the top 15 keywords ranked by tf-idf within the party platforms, treating each topic as a single document. The *parliamentary* and *other* topic keywords were generated by the authors.

Table A1: Topics and keywords used to fit the keyATM model

Category	Keywords
Agriculture	farm, rancher, farmer, ranch, export, commod, agricultur, fiber, grain, embargo, pariti, crop, livestock, food, wheat
Civil Rights	abort, disabl, gender, religi, sex, discrimin, religion, ballot, de-segreg, vote, equal, segreg, reproduct, marriag, racial
Culture	art, artist, endow, film, museum, danc, leisur, opera, orchestra, theatr, scholar, heritag, writer, scholarship, music, cultur
Defense	nato, nuclear, missil, weapon, ballist, veteran, iraq, soviet, treati, troop, korea, allianc, deploy, vietnam, arm
Domestic Commerce	antitrust, merger, mortgag, gambl, dodd, lend, patent, sba, theft, conglomer, ftc, frank, consum, small, loan
Education	student, classroom, teacher, math, tuition, parent, read, academ, graduat, teach, elementari, english, childhood, bilingu, secondari
Energy	oil, gas, coal, solar, energi, nuclear, electr, petroleum, atom, geotherm, opec, decontrol, wind, fossil, ethanol
Environment	speci, pollut, emiss, wetland, superfund, toxic, air, carbon, greenhous, esa, soil, brownfield, wildlif, fish, habitat
Foreign Trade	export, trade, tariff, currenc, negoti, textil, reciproc, monetari, bilater, nafta, china, agreement, protectionist, gatt, foreign
Government Operations	postal, district, columbia, lobbi, census, elector, mail, servant, statehood, ballot, incumb, branch, candid, vote, sunset, usp

Health	medicar, medicaid, patient, hiv, healthcar, drug, coverag, nurs, diabet, mental, cancer, medic, prescript, diseases, health
Housing	homeownership, slum, mortgag, fha, rent, rental, urban, tenant, homeless, rural, fanni, freddi, mac, mae, neighborhood
Immigration	immigr, refuge, undocu, deport, visa, alien, reunif, english, amnesti, newcom, flee, asylum, citizenship, illeg, admiss
International Affairs	israel, africa, soviet, taiwan, palestinian, east, arab, cuba, peac, korea, terrorist, ireland, asia, afghanistan, cuban
Labor	overtim, hartley, taft, pension, bargain, picket, employe, bacon, davi, collect, worker, arbitr, autom, osha, union
Law and Crime	gun, crime, crimin, drug, sentenc, offend, firearm, juvenil, polic, prison, victim, narcot, pornographi, traffick, marijuana
Macroeconomics	deficit, inflat, monetari, bracket, spend, debt, incom, wealthi, wealthiest, recess, taxat, loophol, inflationari, estat, code
Public Lands	puerto, indian, rico, guam, forest, nativ, hawaiian, tribal, virgin, samoa, mariana, tribe, miner, park, wilder
Social Welfare	welfar, parent, needi, nutrit, stamp, social, elder, child, recipi, disabl, lunch, older, charit, mother, poverti
Technology	space, nasa, broadband, internet, broadcast, scientif, telecommun, orbit, saturn, spacecraft, satellit, scienc, cyber, entertain, media
Transportation	highway, railroad, merchant, passeng, rail, freight, airport, transport, mode, maritim, congest, traffic, amtrak, marin, truck
Parliamentary	yield, gentleman, consent, amend, time, minut, senat, hous, bill, order, thank, committe, move, vote, quorum, motion, tabl
Other	peopl, go, get, got, laughter, know, thing, want, say, think, thank

A.2 Pre-Processing

To pre-process our speeches, we first subset to speeches with more than thirty words (following Noble 2024). We then tokenize to unigrams, remove numbers, punctuation, symbols, and separators, lowercase words, remove English stop words from the quanteda package and tokens with fewer than three characters, stem words, and remove words that appear fewer than 20 times or in fewer than 15 documents in each corpus. To match the corpus from Noble (2024), which we use for the 104–114th Congress, we also exclude a small set of phrases including the president of the senate, the president pro tempore, `mr`president, `mister` president, and `madam`[e]? president.

A.3 Topic Validation

To ensure our topic model and subsequent top-1 labels are accurate, we follow Ying, Montgomery and Stewart (2022) in conducting an Optimal Labeling (OL) task with two trained coders. One of the authors randomly sampled 105 speeches from the corpus (five per policy topic) and presented the coders (the other author and a trained research assistant, both blind to the model labeling) with each full speech and four potential labels. One of these was the top label assigned by the model, while the other three were selected from among the remaining possible topics.¹⁴ Coders were asked “What is the main policy category for this speech?” and selected one of the options. The model matched the author and RA in 80% and 81% of cases respectively—accuracy similar to that of Ying, Montgomery and Stewart (2022). Inter-rater reliability between the two coders was substantial, achieving a Cohen’s κ of 0.77.

In Table A2, we provide additional summary statistics on our Optimal Label (OL) topic model validation task.

In Table A3, we break down our OL accuracy by topic category.

¹⁴We exclude the next two most probable topics given that several CAP topics overlap (e.g., macroeconomy and domestic commerce) and could cause confusion.

Table A2: Human Coder Validation Statistics

Statistic	RA	Author
Total Speeches Assigned	105	105
Correct Classifications	85	84
Overall Accuracy (%)	81	80
Inter-Rater Reliability		
Percent Agreement (%)	78.1	
Cohen's Kappa	0.77	
Kappa p-value	< 0.001	

Notes: Validation based on 105 manually coded speeches compared against the keyATM top-1 topic labels. Percent agreement calculated as proportion of speeches where both coders provided identical classifications.

Table A3: Category-Specific Accuracy by Coder

Topic Category	N	RA (%)	Author (%)
Agriculture	5	100	100
Civil Rights	5	100	100
Culture	5	100	80
Education	5	100	100
Environment	5	100	100
Health	5	100	100
International Affairs	5	100	80
Social Welfare	5	100	80
Defense	5	80	80
Domestic Commerce	5	80	80
Energy	5	80	100
Foreign Trade	5	80	60
Government Operations	5	80	80
Labor	5	80	60
Law & Crime	5	80	100
Macroeconomics	5	80	80
Technology	5	80	100
Housing	5	60	40
Transportation	5	60	60
Immigration	5	40	60
Public Lands	5	20	40
Overall Accuracy	105	81	80

Notes: Category-specific accuracy rates for each of the 21 topic categories. Each category contains exactly 5 speeches in the validation sample. Accuracy calculated as the percentage of speeches correctly classified within each category.

A.4 Construct Validity: Embedding-based Similarity Validation

A.4.1 Comparsion Sampling

We sampled focal speeches and paired comparison texts across our time series subject to a few constraints. To validate within-focal speech similarity scores, not just across-focal speech similarity, we ensured that each focal speech appeared multiple times with different sets of comparison speeches. We also wanted to assess whether our RA could discriminate between speech pairs that were more or less similar. To that end, we calculated the standard deviation of OpenAI-3S, BGEM3-ColBERT, and BGEM3-dense scores across our corpus and then determined, for each focal speech, how large was the difference of similarity scores between each of the comparison speeches. We grouped comparison tasks into buckets based on the size of this difference, such as below 0.25 of a standard deviation, 0.25–0.50 of a standard deviation, up to greater than 2 standard deviations. We sampled approximately 30 comparison tasks per bucket for a total of 250 tasks.

A.4.2 Example Task

Below are the instructions we provided to the RA in how to think about similarity when selecting speeches:

Your task is to help validate an automated measure of similarity between political speeches. Below, you'll be shown a series of sets of two **comparison speeches**. For each pair of comparison speeches, choose the one that you believe is **more similar** to the **focal speech** above.

How to think about similarity:

Similarity in this context is **broad and multidimensional**. There is no single correct criterion. Instead, consider the following dimensions when making your judgment:

- **Topic:** Are the speeches about the same issue or subject?
- **Position:** Do the speakers take similar stances or make similar arguments?
- **Language:** Do they use similar words or phrases?
- **Tone/Sentiment:** Is the emotional tone or attitude similar (e.g., optimistic, angry, urgent)?
- **Style:** Do they share rhetorical features like repetition, emphasis, or structure?

You do **not** need to agree with the speaker or analyze the truth of the content—just assess overall similarity in meaning, purpose, and presentation.

There will often be some overlap across both comparison speeches. Use your best judgment to select the one that feels closer **overall** to the focal speech. There are no trick questions—your intuitive assessment is valuable for this task.

Please note: sometimes, you will encounter a repeated comparison speech paired with a different comparison speech. That is expected. Just keep choosing the most similar one.

Then, the RA was presented with a series of tasks like the one below.

Primary Speech (1120072647): mr. chair. i rise today in strong opposition to the socalled "workforce democracy and fairness act" . the changes to union election procedures promoted in this bill are the exact opposite of the kind of fair and democratic policies that our working families need. instead of focusing on job creation and the revitalization of our middle class. the republicans in this chamber are once again promoting legislation that undermines the rights of american workers. this proposed legislation would limit the ability of the national labor relations board to interpret our nations labor laws and to protect workers right to unionize. for over 75 years. the national labor relations act has guaranteed the rights of employees to organize and bargain collectively. or to refrain from such activity if they choose. during the new deal. our predecessors in this body created the national labor relations board as an independent agency charged with the oversight and enforcement of these rights. h.r. 3094. which overturns the rulings of the nlr. undermines its charge to maintain fair and democratic relationships between unions and employers. this legislation allows the problem of prolonged delays in union elections to continue unchecked by adding mandatory and arbitrary waiting periods. it seizes from workers the right to determine their own representative membership groups. which would allow unscrupulous businesses to suppress election drives and vote down union representation. it would also make it possible for irresponsible and frivolous litigation to endlessly delay the election process. effectively barring workers from their fundamental right to collective bargaining representation in the workplace. supporting and

protecting americas workers is an essential part of rebuilding our economy and ensuring that all families and communities share in our nations prosperity. our middle class was built on the rights and safeguards that labor unions fought to obtain. from the 40 hour workweek to ending child labor. union representation has helped to guarantee rights that many of us take for granted today. unions negotiate for safe working conditions. living wages. and basic benefits that impact all workers. efforts to decrease the power of collective bargaining in this country in recent decades have been accompanied by an erosion of workers benefits and greater income inequality. this year in wisconsin and ohio. we have seen voters reject recent attempts to strip away the rights of government workers. and we should likewise reject this attempt to limit access to these rights for those in the private workforce. this bill does nothing to protect and support working families. and i urge my colleagues to stand up for workers rights and oppose this bill.

Comparison A (1120072704): i yield myself such time as i may consume. what i say to my good friend from south carolina is that i have the greatest respect for employers. id like the gentleman to join me in passing the american jobs act to give them payroll tax relief and to give them tax credits for hiring new employees. but you have to ask the question: after this bills implementation. will workers view their workplaces more favorably? will their wages match the growth rates of the companies and economy? will workers feel like american employers. supported by government. provide meaningful safety for community survival? this legislation. frankly. undermines the american workers. can we all get along? can we find a way to address the concerns of making sure that we are fair to the employer but not have delay after delay after delay to deny someone his constitutional right of organizing freedom of expression? i think we can. the elimination of the provisions that i have spoken of is a dilatory upper hand of employers to get the better hand of our employees. i reserve the balance of my time. the acting chair. the gentlewoman from texas has 15 seconds remaining. and the gentleman from south carolina has 45 seconds remaining.

Comparsion B (1120072655): mr. chair. i rise today in opposition to h.r. 3094. the republican plan to crush workers rights and destroy any glimmer of hope our working families have at economic recovery. the republicans designed this bill to destroy 75 years of national labor review board case law in their attempt to

dismantle the middle class. collective bargaining and the right to organize helped build a strong american middle class. it doesnt cost the federal government one dime in real money. instead of taking steps to create jobs and strengthen working families. republicans are dismantling key worker protections. all workers should have the ability to negotiate with their employer about salary and benefits. whether theyre in a union or not. organized labor is great for business. thousands of companies across the country thrive with a unionized workforce. those businesses recognize that their employees deserve to have a safe workplace and fair wages and benefits. thats just good business. this bill encourages corporations to stall nlr elections while they mount a onesided. antiunion campaign. at its core. this is an undemocratic bill that undermines our values. we have a long established process for workers to attempt to form a union and collectively bargain with employers. employers and employees should stay on equal ground in the process. there is no need to deny workers their right to a free and fair union election. many of my republican friends like to talk about the issue of tort reform. they like to tell us that we have to prevent frivolous lawsuits they cost taxpayers millions and millions of dollars and they drag down the economy. i have news for my republican friends: the election prevention act encourages frivolous litigation. this bill will mean mountains of litigation before union elections can be held. the result is a massive backlog. guess who picks up the tab? the american taxpayer! we have important issues facing our country and it boggles my mind that we are taking up yet another bill that does nothing to get our friends and neighbors back to work. we need to focus on lowering the unemployment rate and creating jobs not taking away the rights of hardworking americans. i urge my colleagues to recognize this veiled attempt to destroy the rights of american working families.

Which comparison speech (A or B) is more similar to the primary speech?

A: 1120072704

B: 1120072655

Here, the RA and embedding-based choice was comparison B. As can be seen on a careful reading, comparison B is more similar to the focal speech for several reasons. First, although all speeches concern relationships between capital and labor, the focal speech and speech B explicitly discuss the NLRB whereas comparison A does not. Both the focal speech and comparison B also take a clear pro-labor position, whereas comparison A

takes a more balanced approach favoring both employers and labor. Both the focal speech and comparison B are emotional, making use of hyperbole and extreme claims such as “irresponsible and frivolous litigation to endlessly delay” (focal speech) and “destroy any glimmer of hope our working families have” (comparison B). Comparison A makes use of a different rhetorical strategy, questioning. For these reasons, it is clear why both the RA and model selected comparison B.

A.4.3 Bucket Accuracy

In Table A4, A5, A6, we break down the accuracy of comparison tasks by standard deviation bucket. Within a comparison task, we determine how close these two pairwise differences are as compared to the standard deviation of differences across our full dataset. For example, if the pairs within a comparison task have an OpenAI-3S cosine similarity difference of less than 0.035, this is a very small difference and they fall into the first “bucket” in the first table.

This table demonstrates that generally, as the difference between the two pairs grows, the RA and embedding-based similarity measures are more likely to select the same comparison speech. When differences are small, accuracy decreases. Such a pattern is more obvious for OpenAI-3S and BGEM3-dense scores. Overall, there appears to be a threshold (around 0.75–1 sd) above which the model is more likely to agree with human judgment. These results suggest that humans may struggle to interpret small differences between speeches that our embedding method captures. This could result from either the embedding method detecting imperceptible differences or error in the machine measurement based on differences that are not relevant when humans process similarity.

Table A4: OpenAI-3S and RA Agreement Rates by Standard Deviation Bucket

Std. Dev. Bucket	Num. Responses	Agreement Rate (%)
0 to 0.25	32	65.6
0.25 to 0.50	30	70.0
0.50 to 0.75	24	70.8
0.75 to 1.00	25	76.0
1.00 to 1.25	29	82.8
1.25 to 1.50	23	82.6
1.50 to 1.75	29	86.2
1.75 to 2.00	28	92.9
> 2.00	30	100.0

Table A5: BGEM3-ColBERT and RA Agreement Rates by Standard Deviation Bucket

Std. Dev. Bucket	Num. Responses	Agreement Rate (%)
0 to 0.25	41	70.7
0.25 to 0.50	37	67.6
0.50 to 0.75	36	61.1
0.75 to 1.00	32	90.6
1.00 to 1.25	22	86.4
1.25 to 1.50	20	95.0
1.50 to 1.75	16	93.8
1.75 to 2.00	14	100.0
> 2.00	32	96.9

Table A6: BGEM3-Dense and RA Agreement Rates by Standard Deviation Bucket

Std. Dev. Bucket	Num. Responses	Agreement Rate (%)
0 to 0.25	39	53.8
0.25 to 0.50	31	54.8
0.50 to 0.75	27	66.7
0.75 to 1.00	34	76.5
1.00 to 1.25	24	91.7
1.25 to 1.50	18	94.4
1.50 to 1.75	32	100.0
1.75 to 2.00	11	100.0
> 2.00	34	91.2

B Additional Empirical Models

B.1 Southern Democrat Application

Table B1: Factional Rhetorical Cohesion, 1961–2020

	(1)	(2)
Southern Democrat Pair	−0.009*** (0.001)	−0.078*** (0.013)
Democrat-Southern Democrat Pair	−0.013*** (0.001)	−0.108*** (0.007)
Republican Pair	−0.012*** (0.001)	0.051*** (0.006)
Congress Number		0.002*** (0.000)
Southern Democrat Pair x Post-1994	0.002 (0.002)	
Democrat-Southern Democrat Pair x Post-1994	0.010*** (0.001)	
Republican Pair x Post-1994	−0.005*** (0.001)	
Southern Democrat Pair x Congress		0.001*** (0.000)
Democrat-Southern Democrat Pair x Congress		0.001*** (0.000)
Republican Pair x Congress		−0.001*** (0.000)
Lawmaker Dyad Fixed Effects	✓	✓
Num.Obs.	4658802	4658802
R2 Adj.	0.061	0.045
R2 Within Adj.	0.004	0.017

Note: Results are from ordinary least squares models. Standard errors are clustered at the lawmaker-dyad level.